

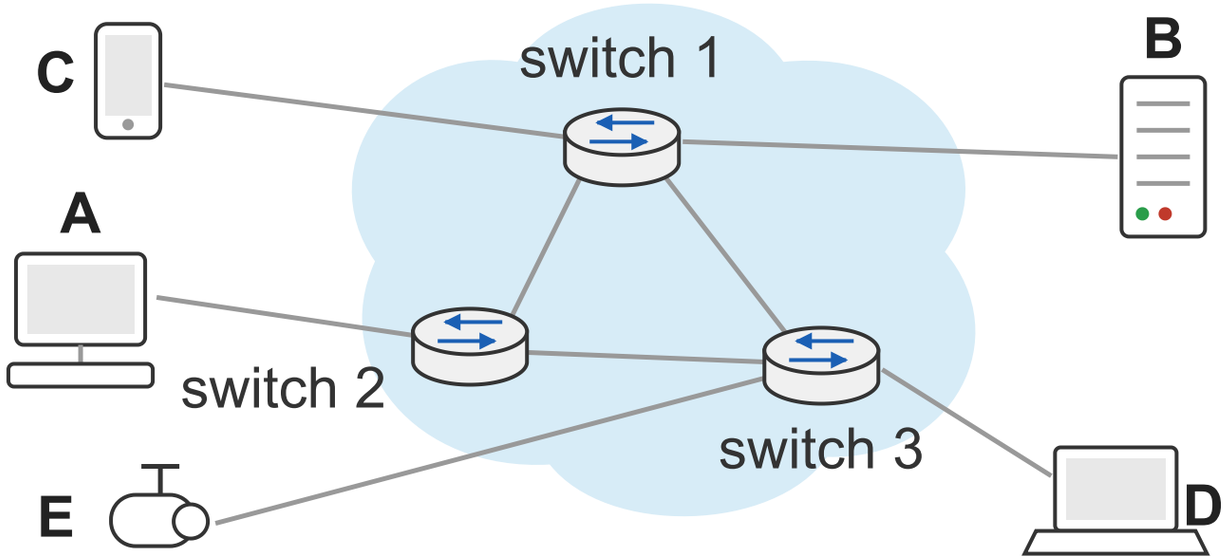
UNIVERSITY OF WATERLOO

Finding paths - Part I

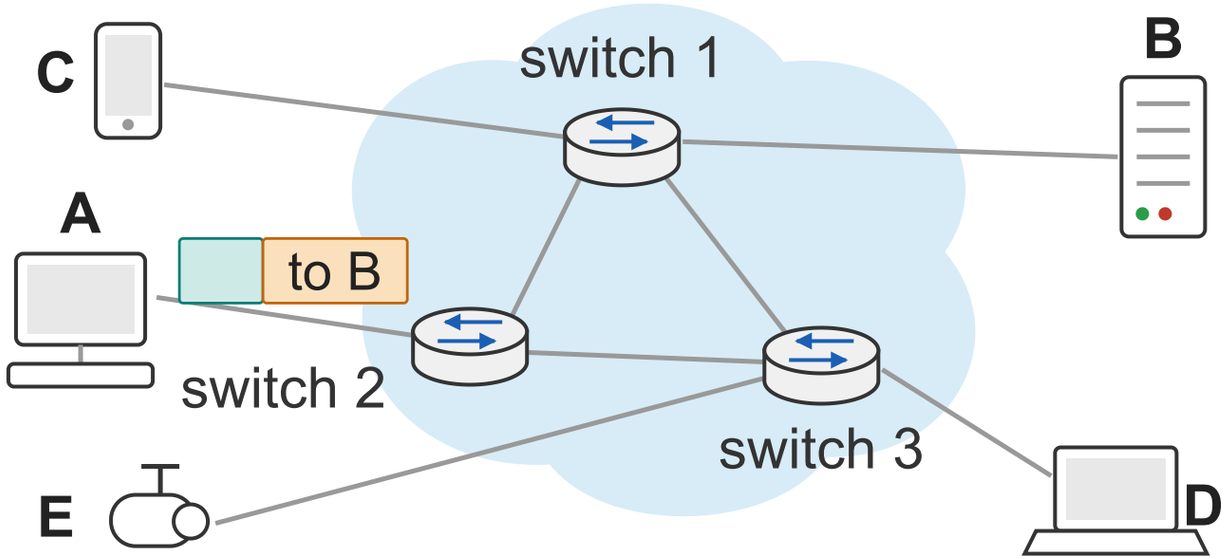
CS 456/656, Fall 2026

Mina Tahmasbi Arashloo

How does the network get a packet from computer A to computer B?

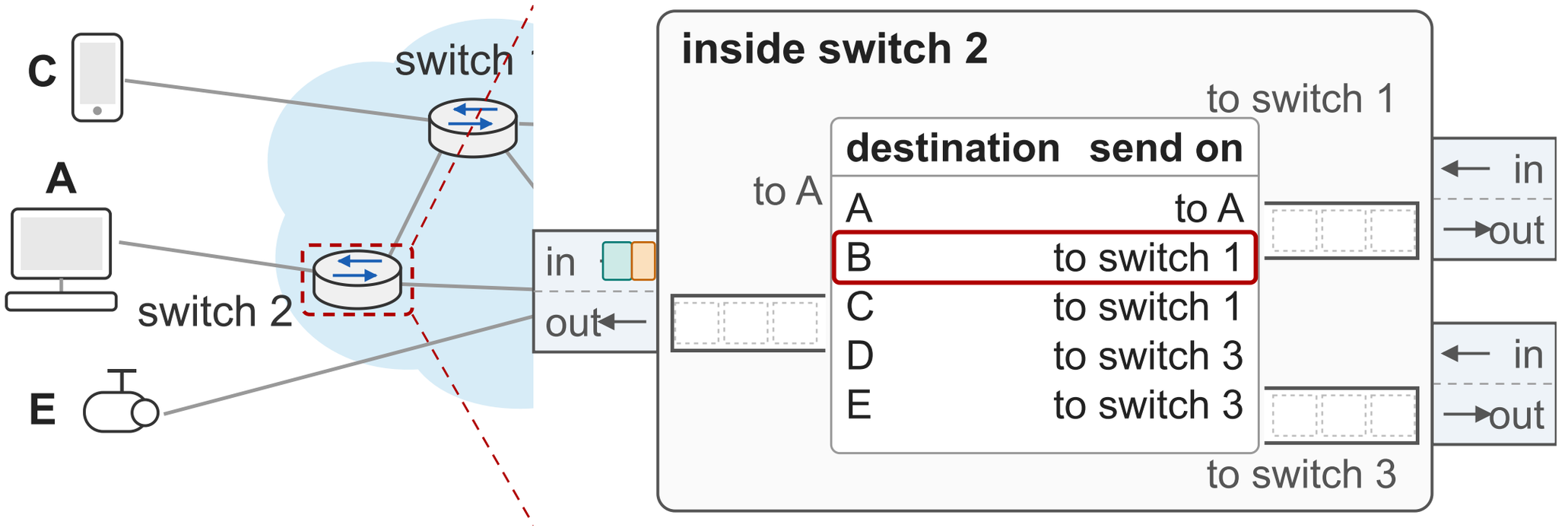


How does the network get a packet from computer A to computer B?



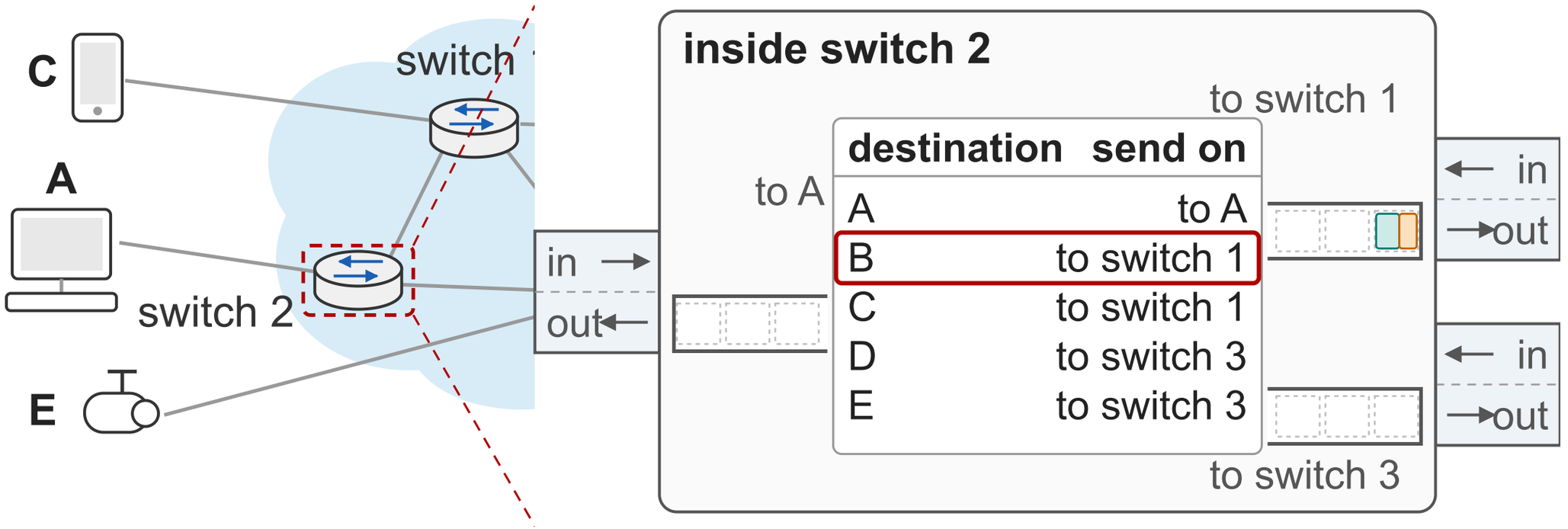
A sends a packet to B over its link to switch 2

How does the network get a packet from computer A to computer B?



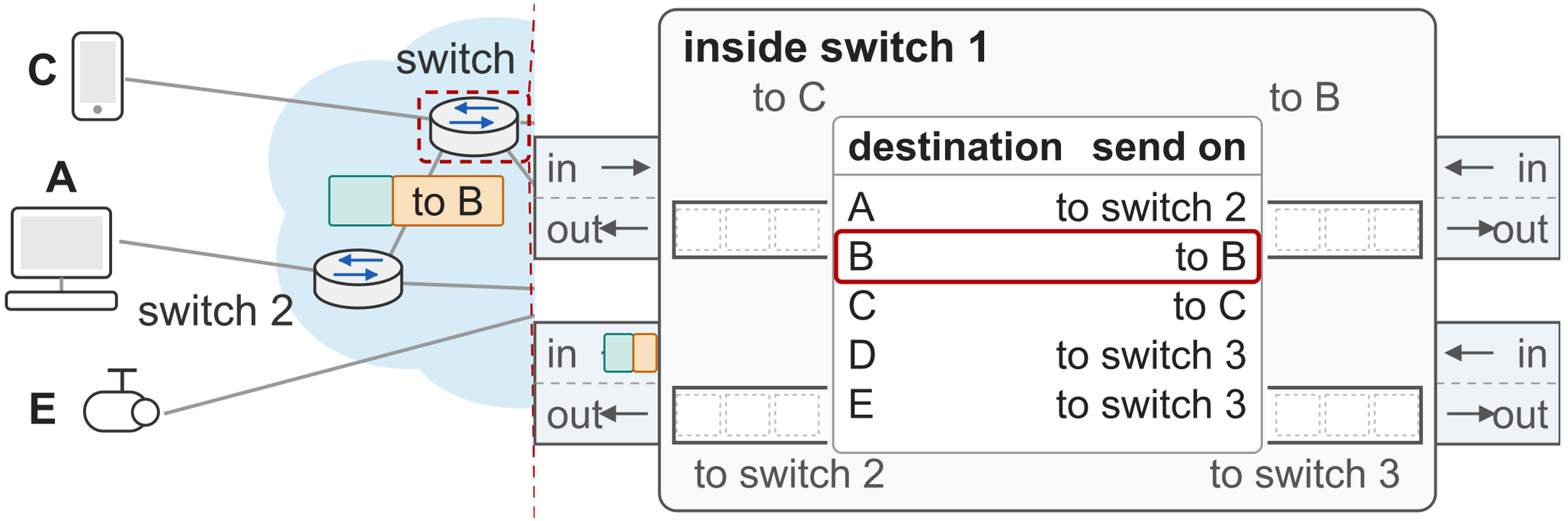
Switch 2 looks up B in its forwarding table.
Sends the packet on the link to switch 1

How does the network get a packet from computer A to computer B?



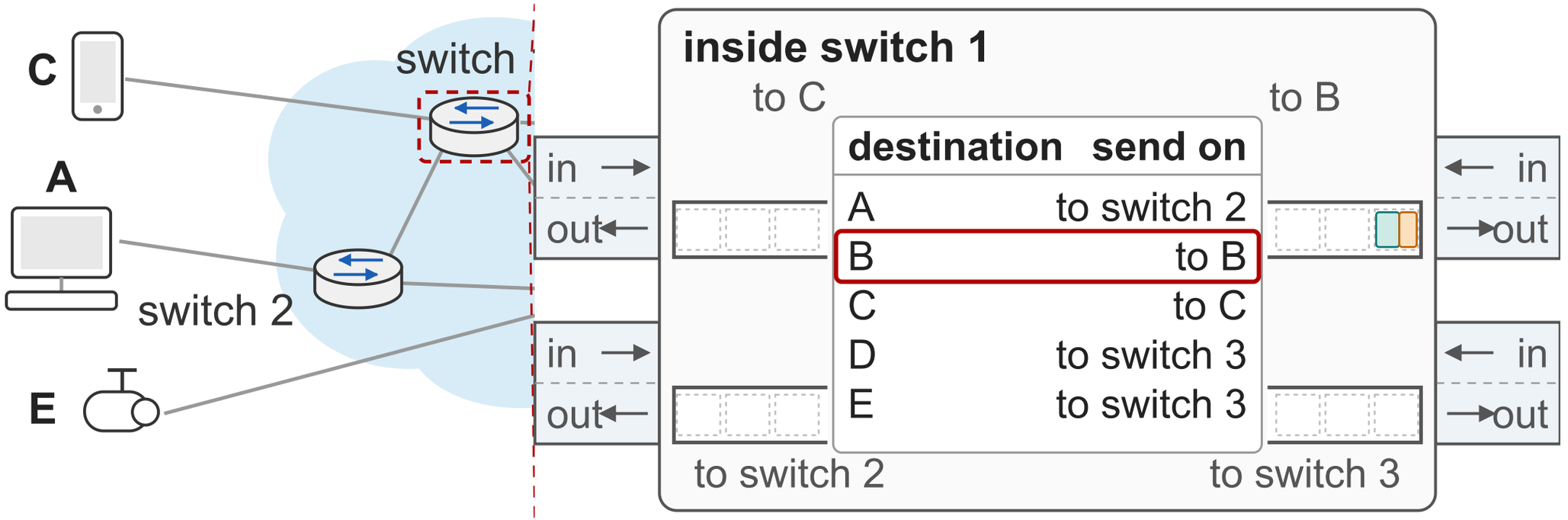
The packet is buffered in that link's queue
to wait for its turn to be transmitted

How does the network get a packet from computer A to computer B?



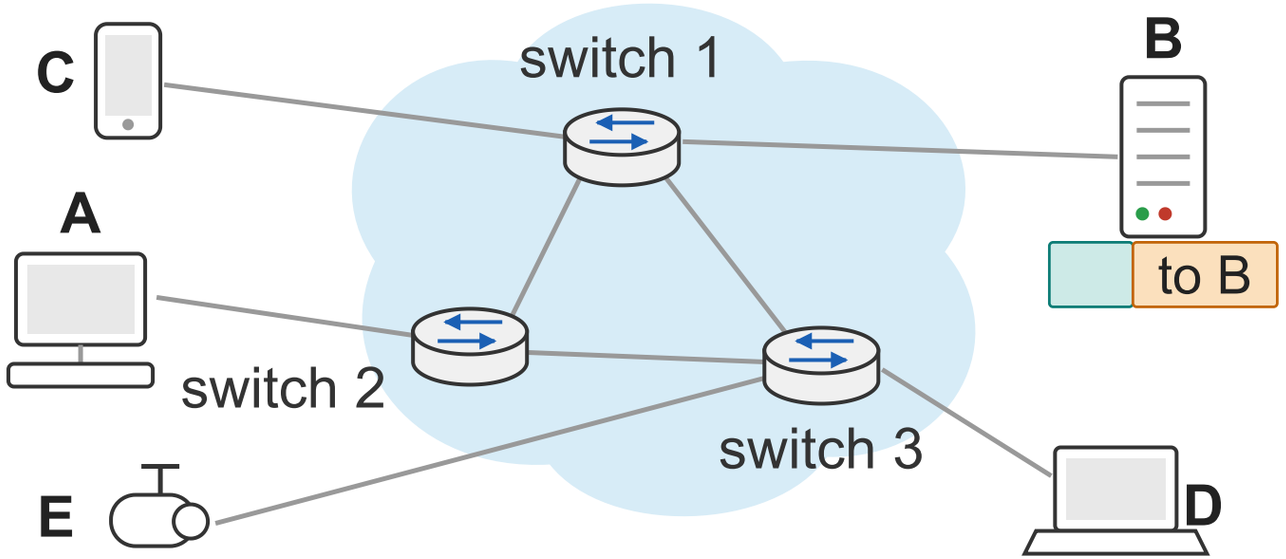
Switch 1 looks up B in its forwarding table.
Sends the packet on the link to B

How does the network get a packet from computer A to computer B?



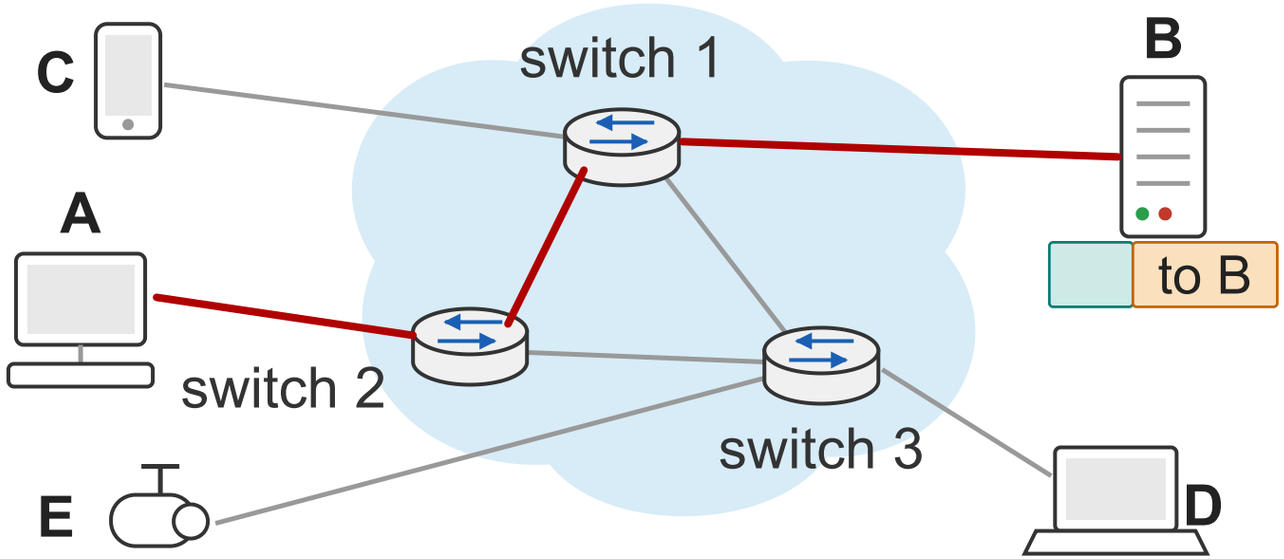
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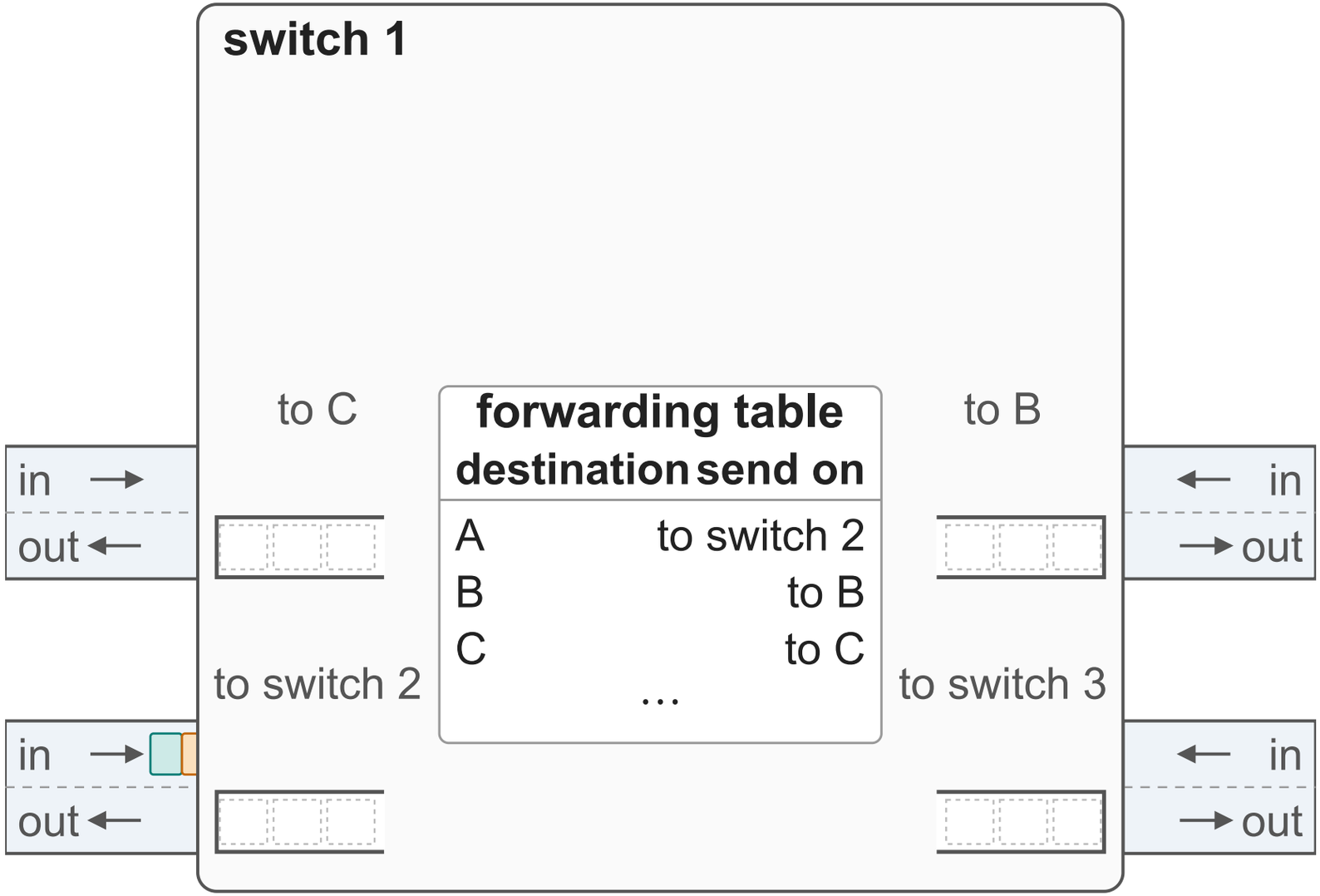
B receives the packet

How does the network get a packet from computer A to computer B?

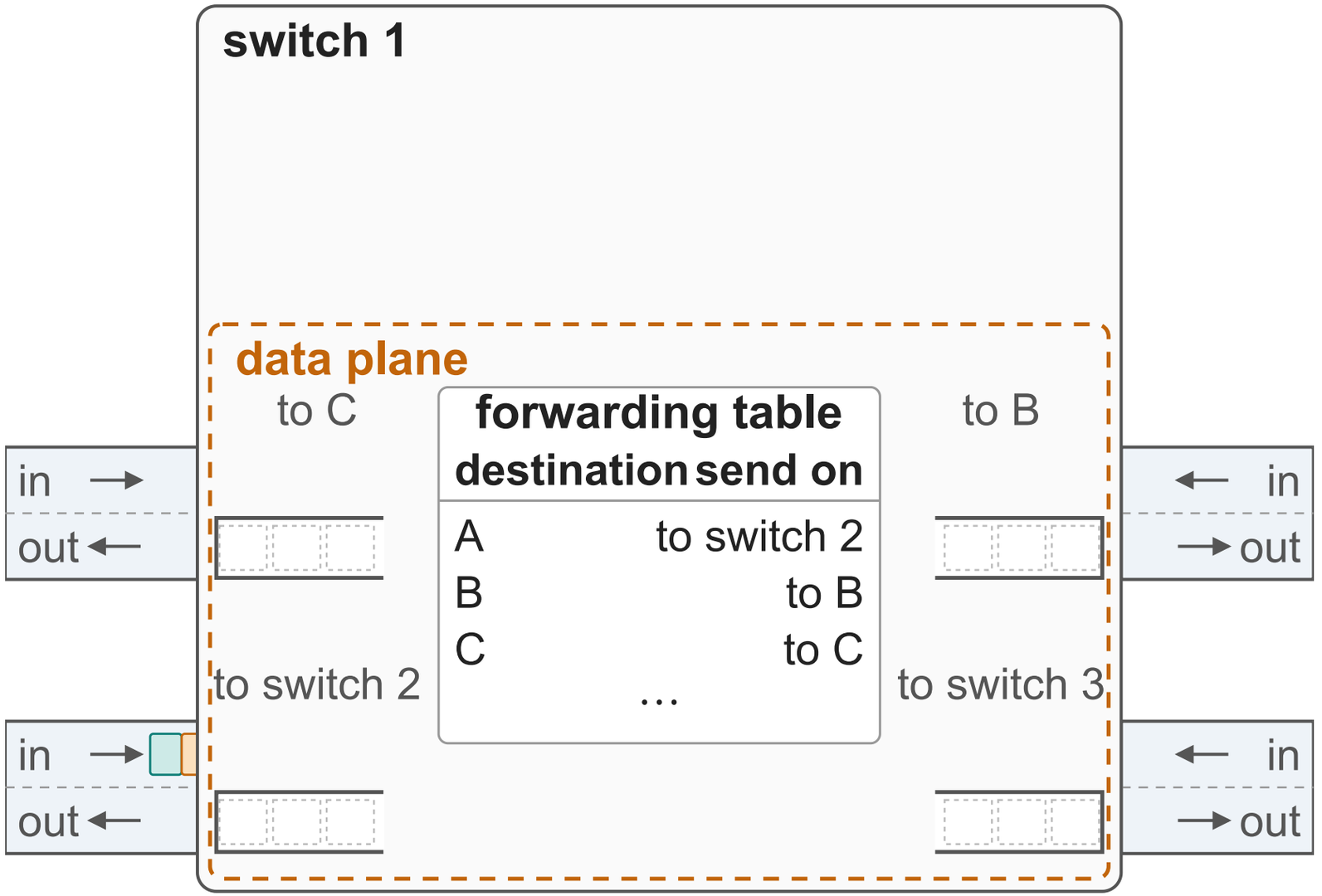


How did the switches know how to forward the packet along the path A → switch 2 → switch 1 → B?

Data plane and control plane



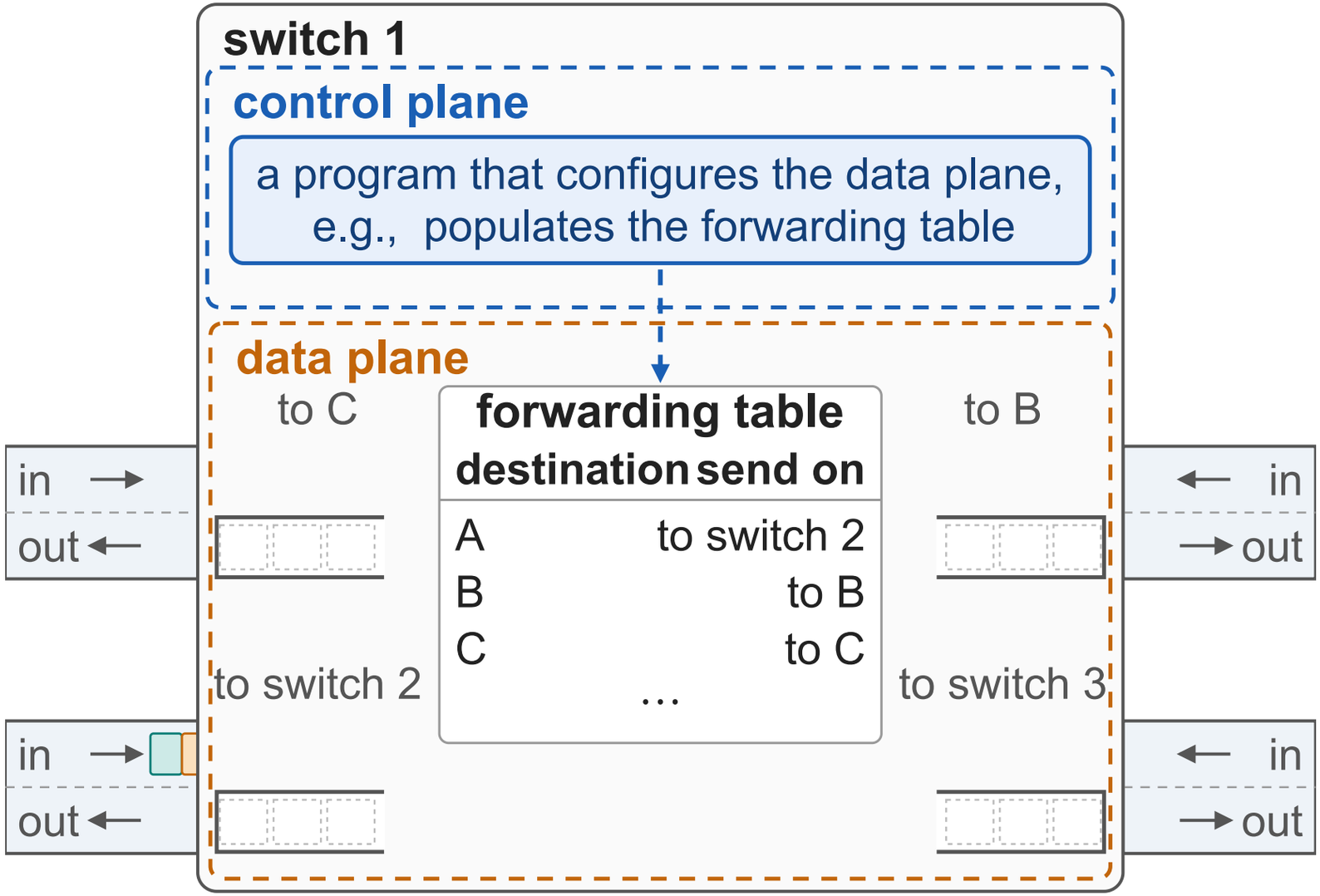
Data plane and control plane



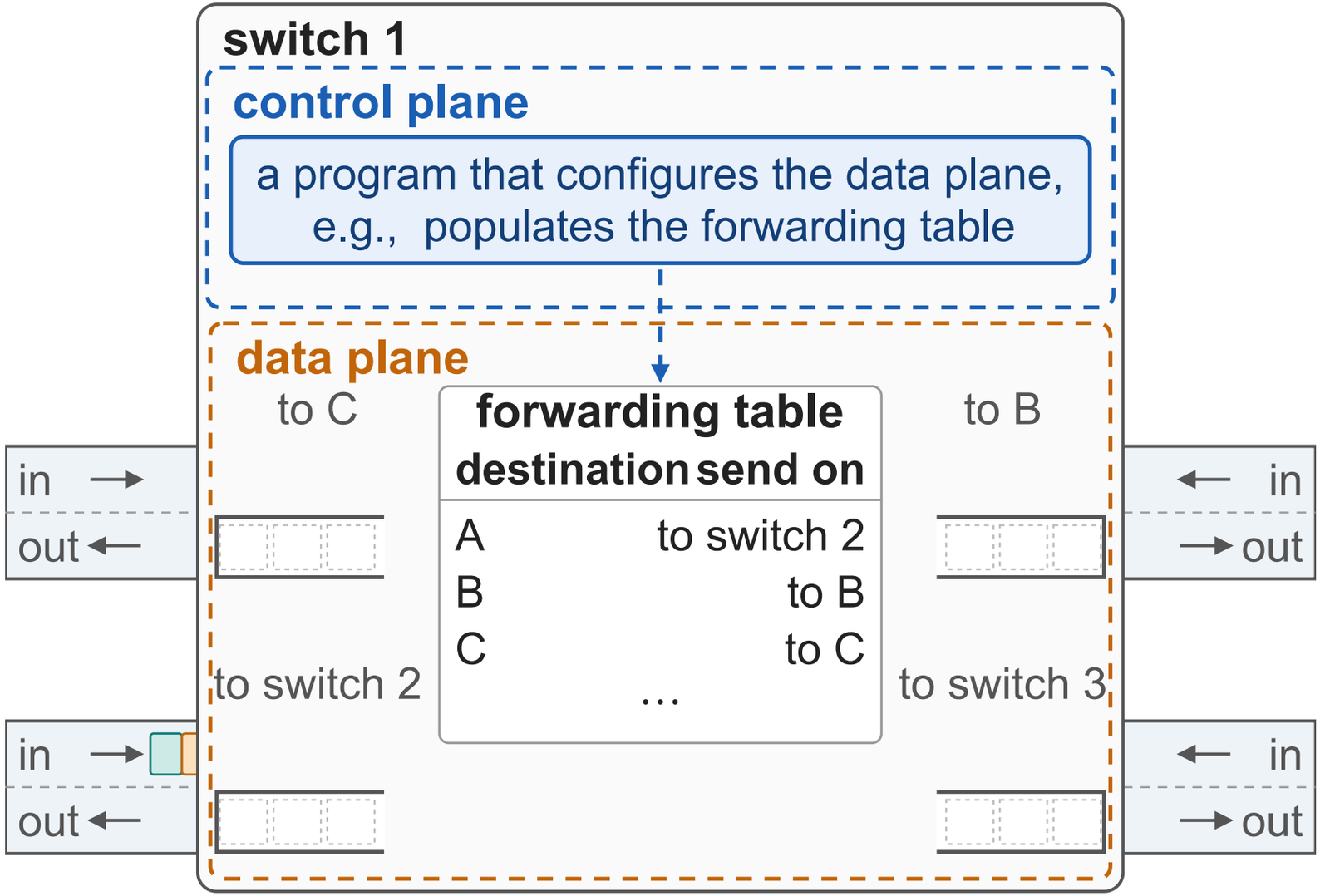
Data plane

- Handles each packet
- One main job: forwarding
  - look up the next hop
  - buffer the packet
  - send it out of the link to next hop
- Has to be fast:
  - On a 100 Gbps link, output a 1000 B packet every 80 ns

Data plane and control plane

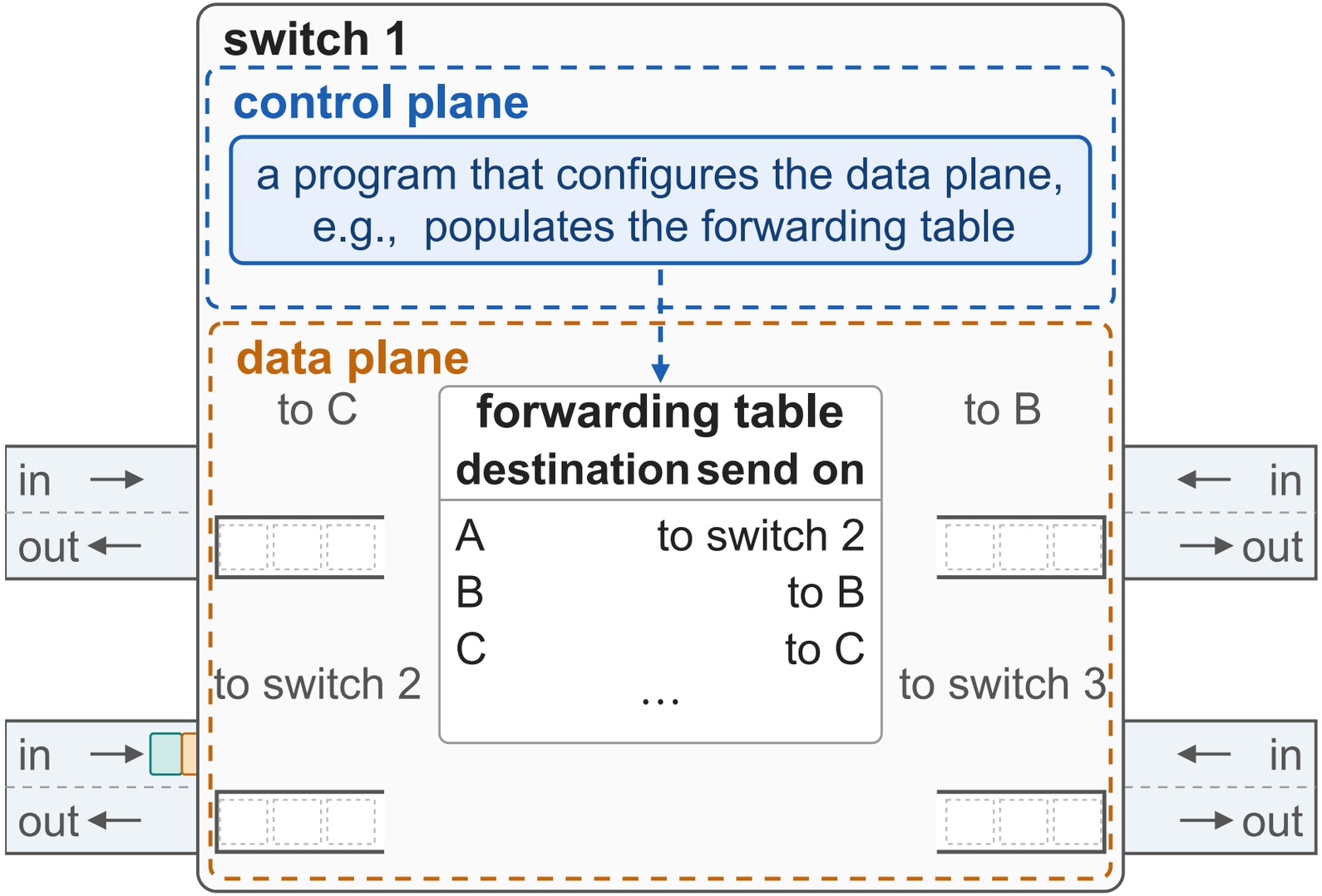


Data plane and control plane

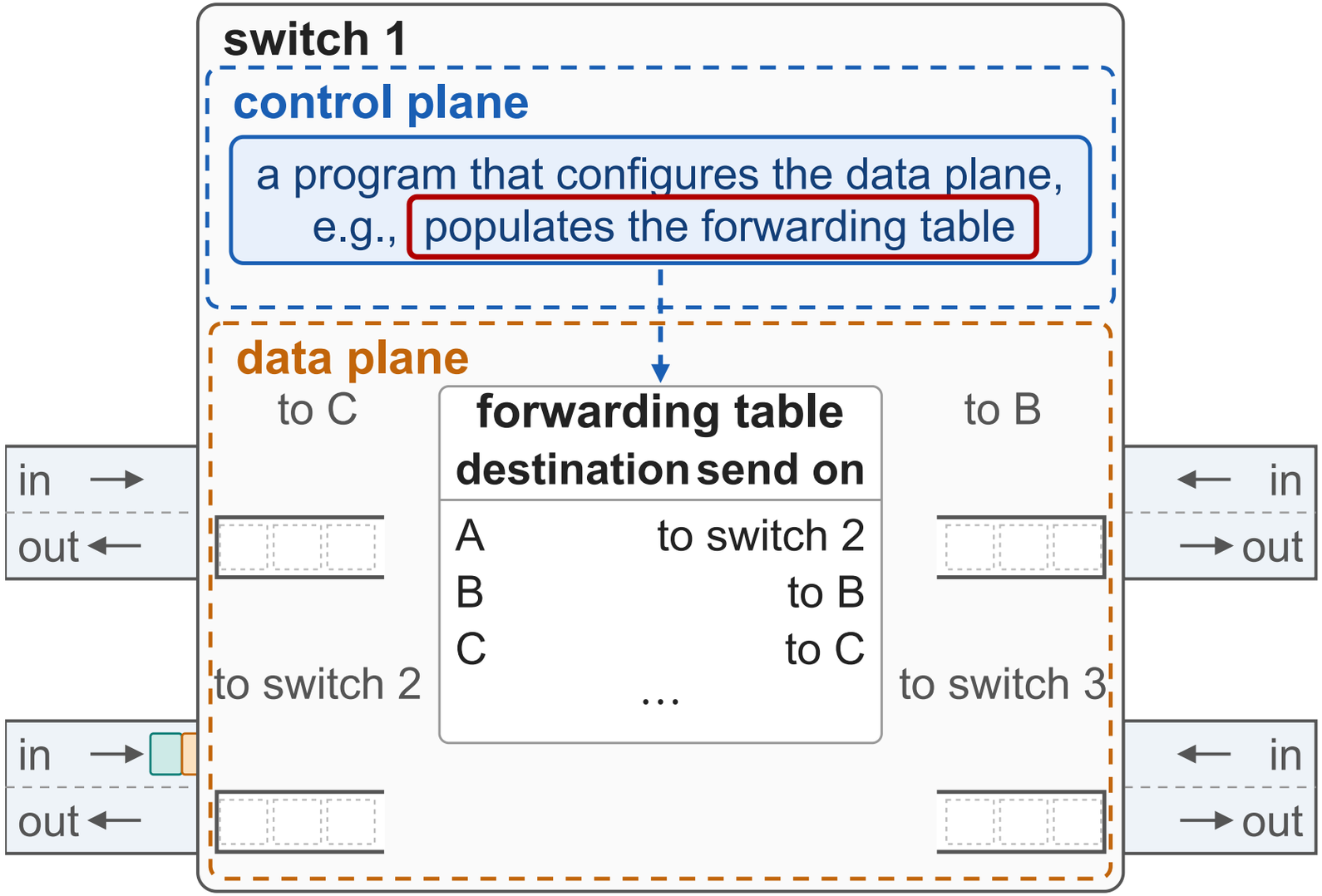


This is a recurring pattern in systems: a fast data path and a slower control path that configures it or handles exceptions.

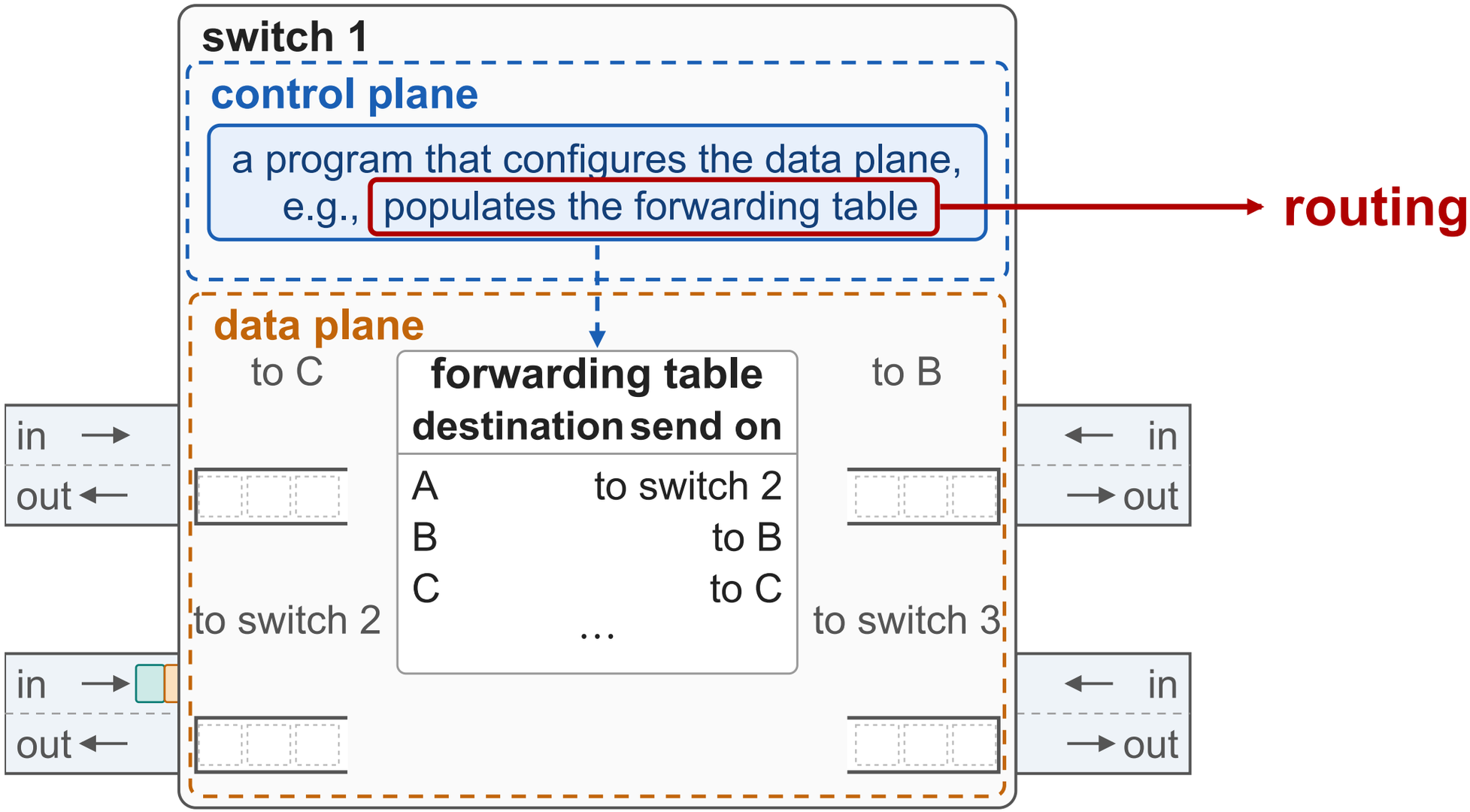
Data plane and control plane



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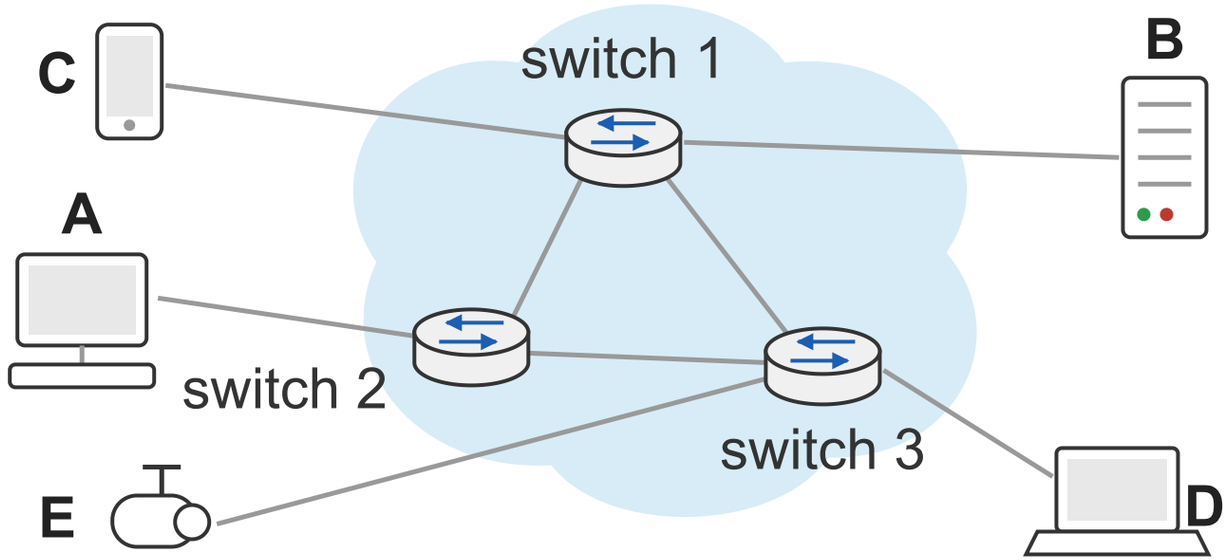
Routing

- One of the main jobs of the control plane is **routing**
- Routing ensures the forwarding table in each switch is populated in a way such that:
  - Every endpoint is *reachable* from every other endpoint
  - over “best” path

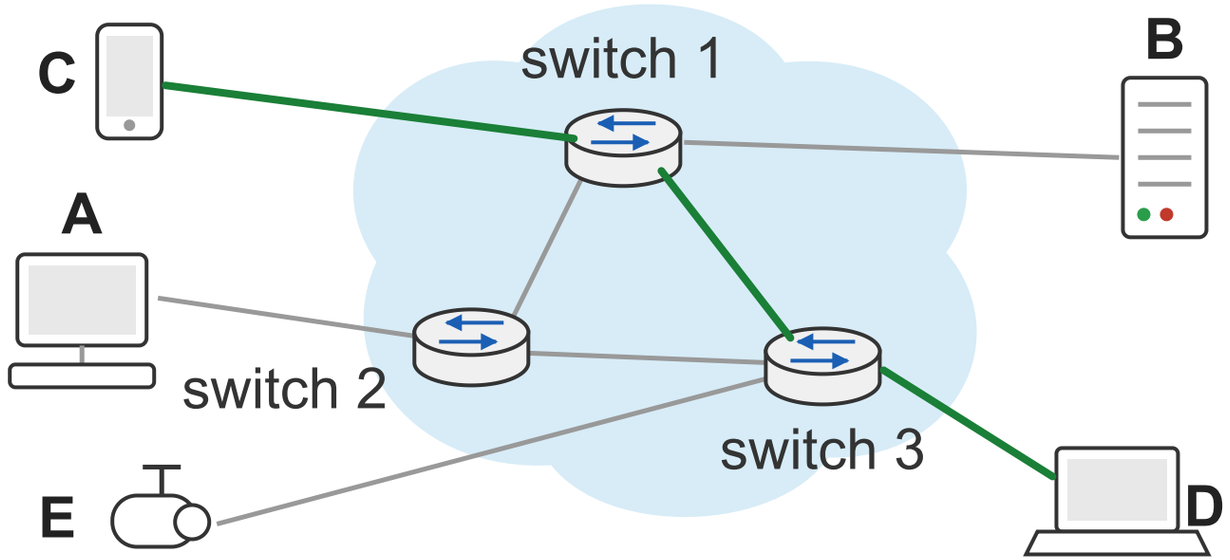
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Fewest hops?

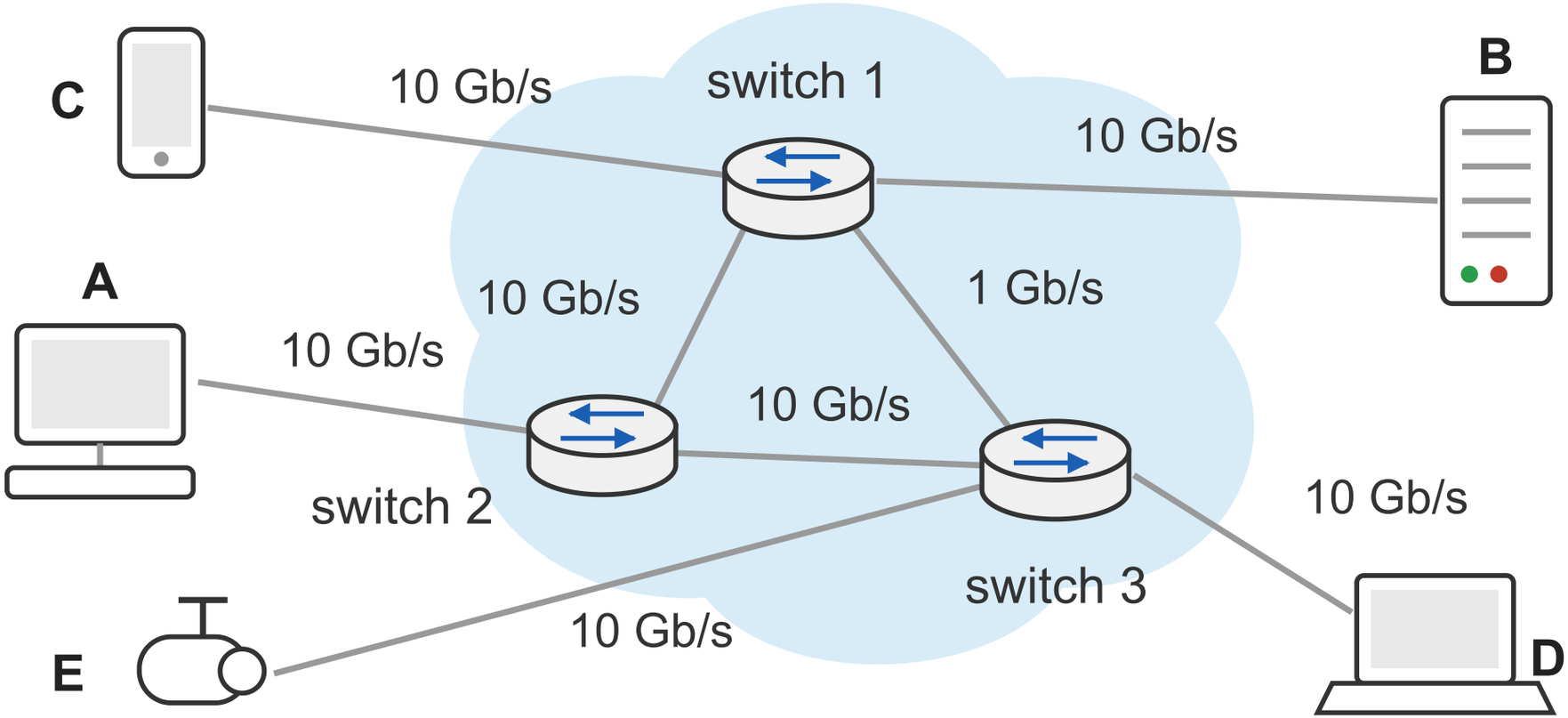


Fewest hops?

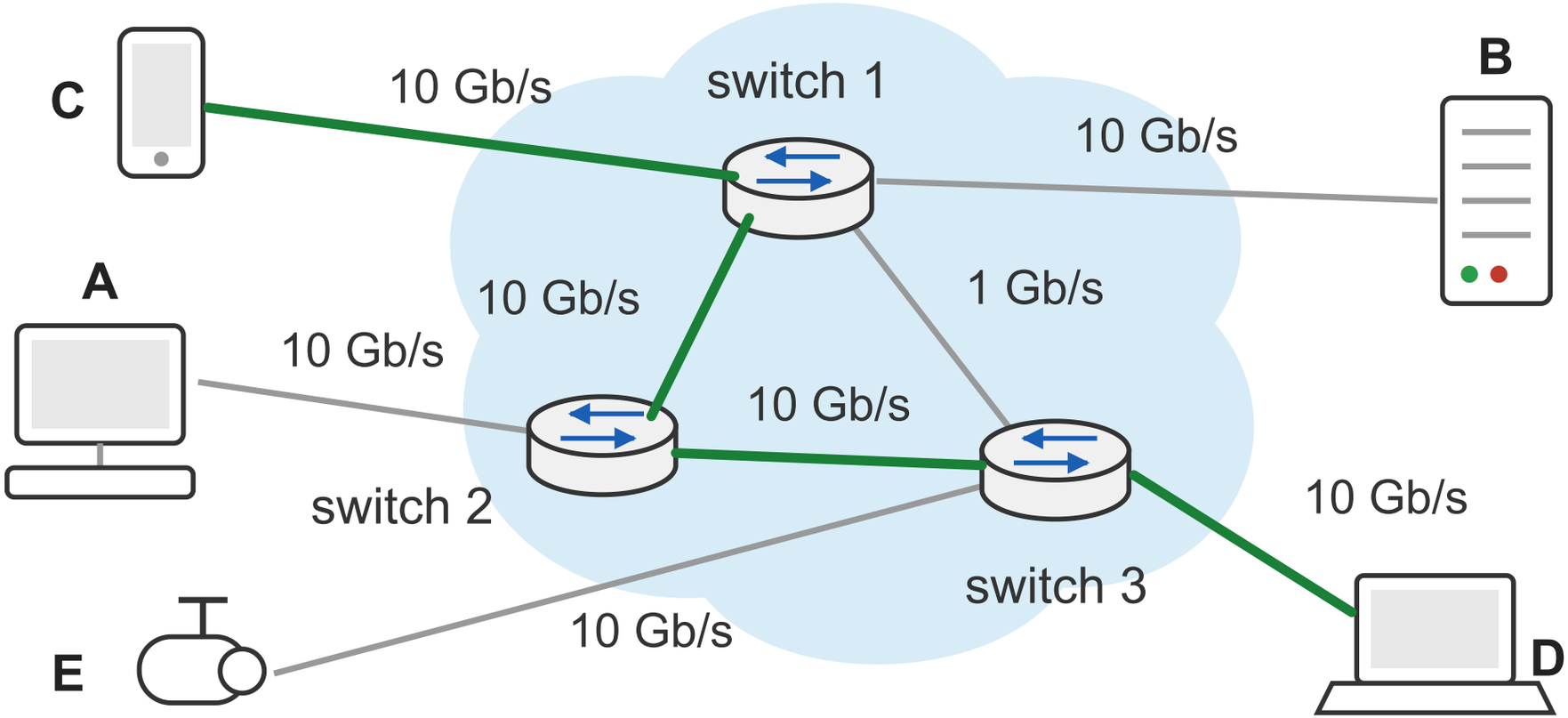


When would you consider this the “best path”?

High throughput?



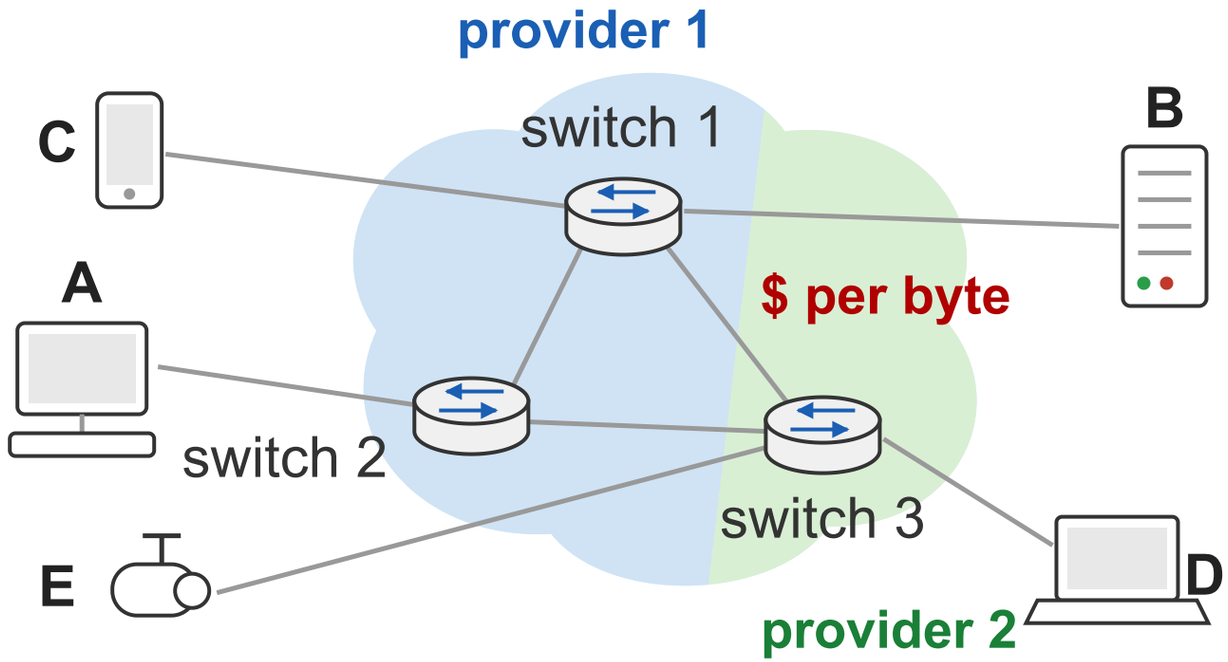
High throughput?



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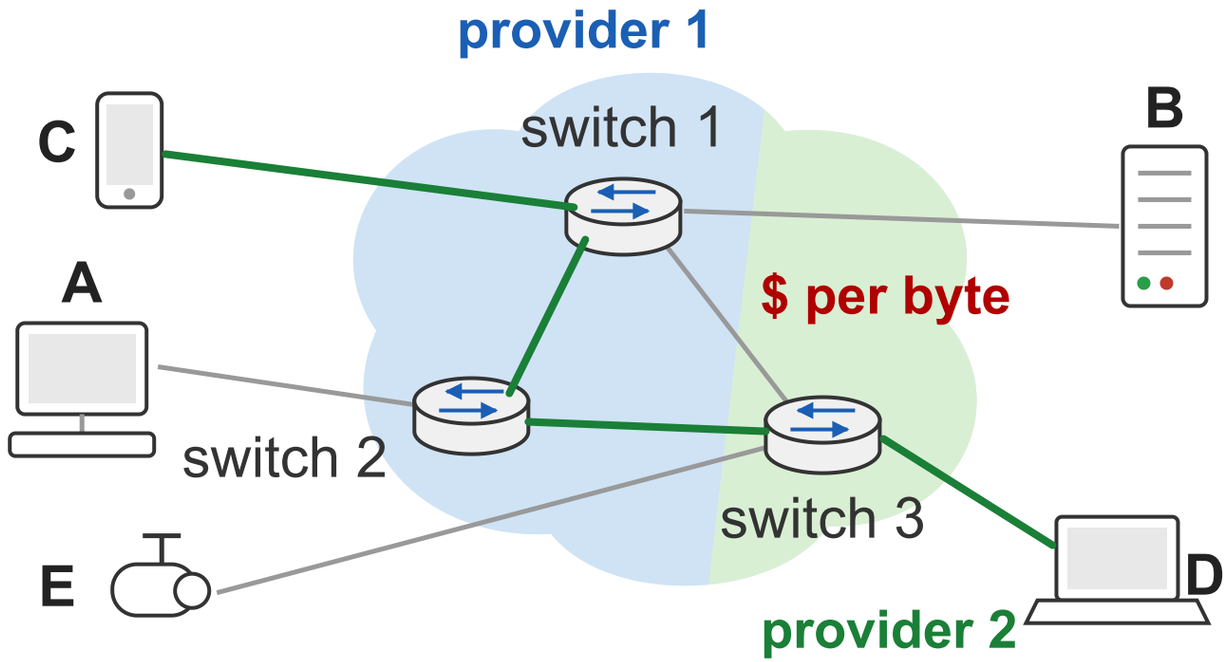
The cheapest?

Two providers. Provider 1 pays provider 2 a flat rate, except on the link from switch 1 to switch 3, billed per byte.



The cheapest?

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Routing

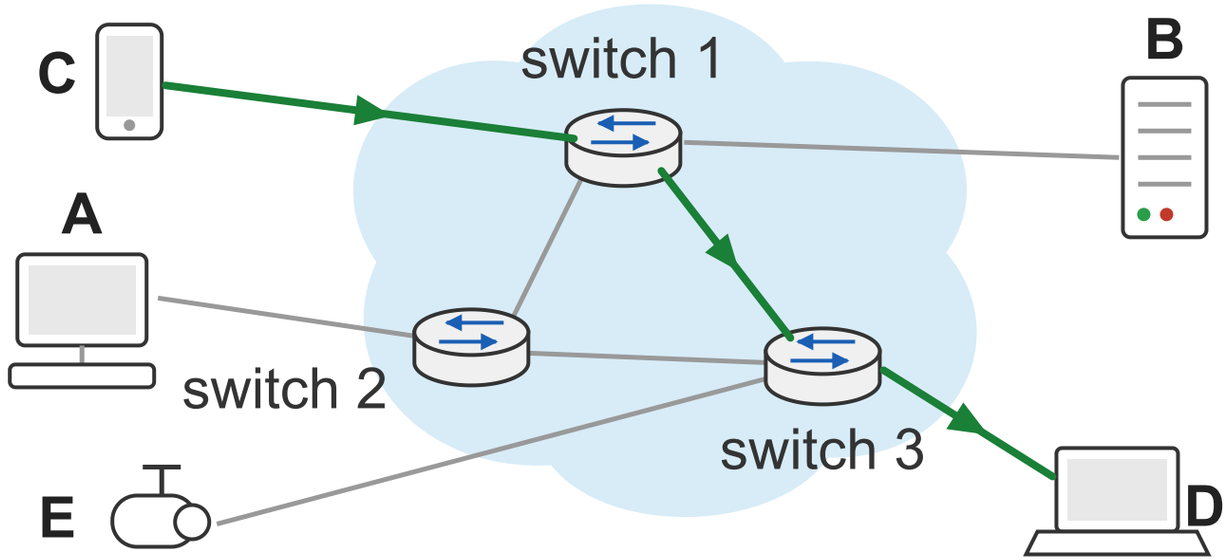
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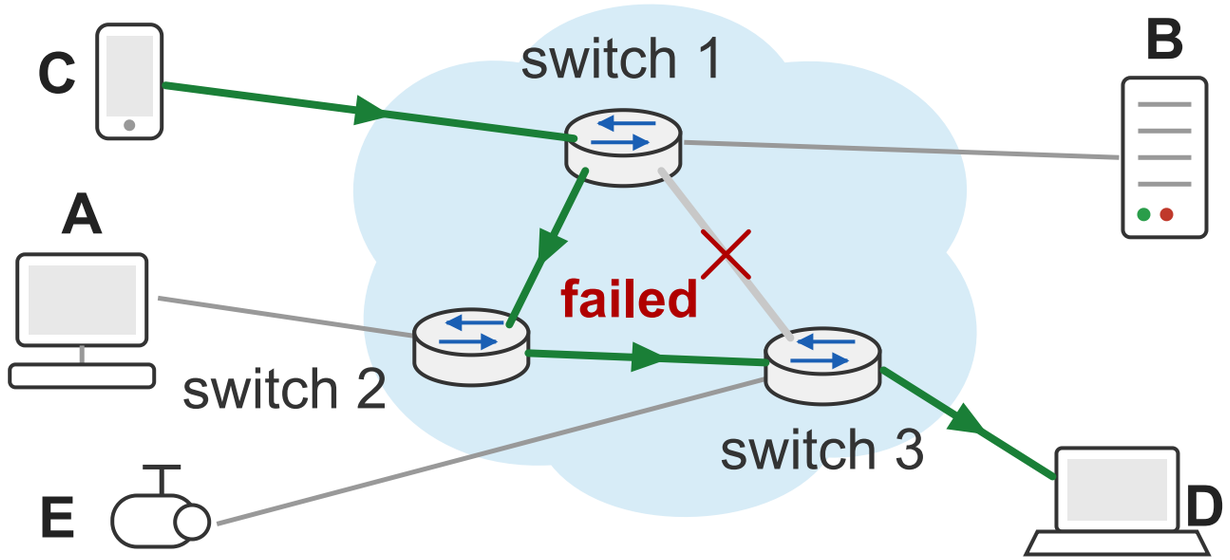
There is no one definition of "best" path.

Routing is not a one-time decision



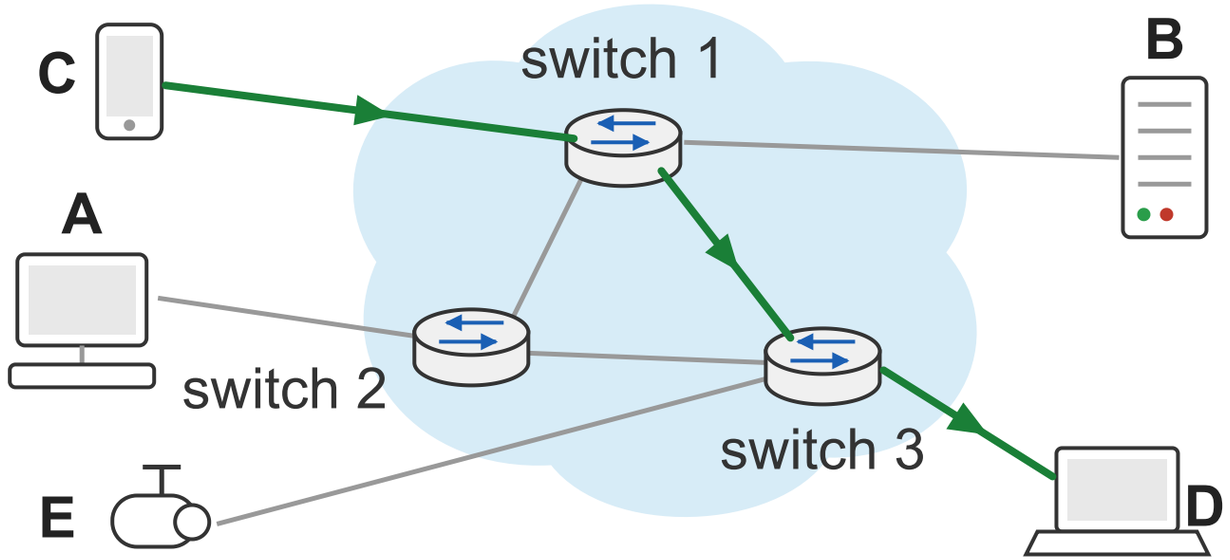
Links can fail

Routing is not a one-time decision



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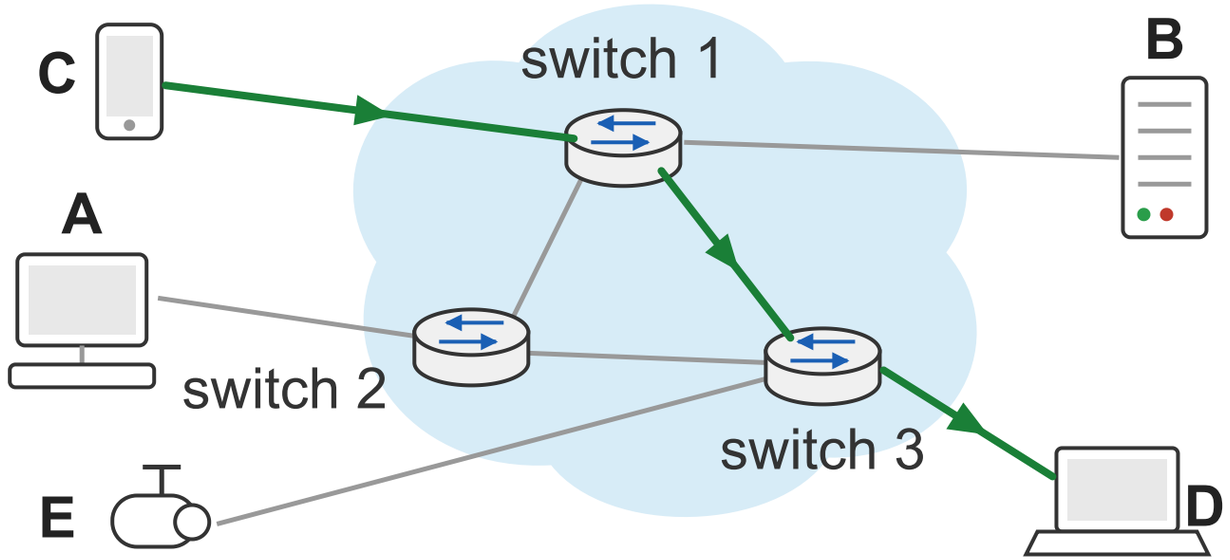
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Links can fail

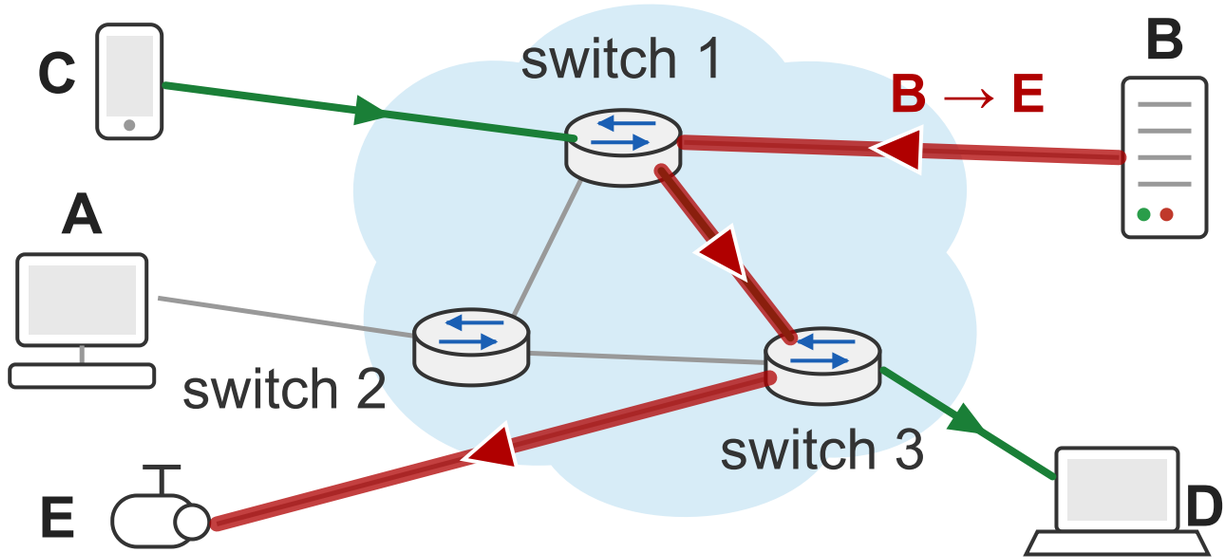
and come back

Routing is not a one-time decision



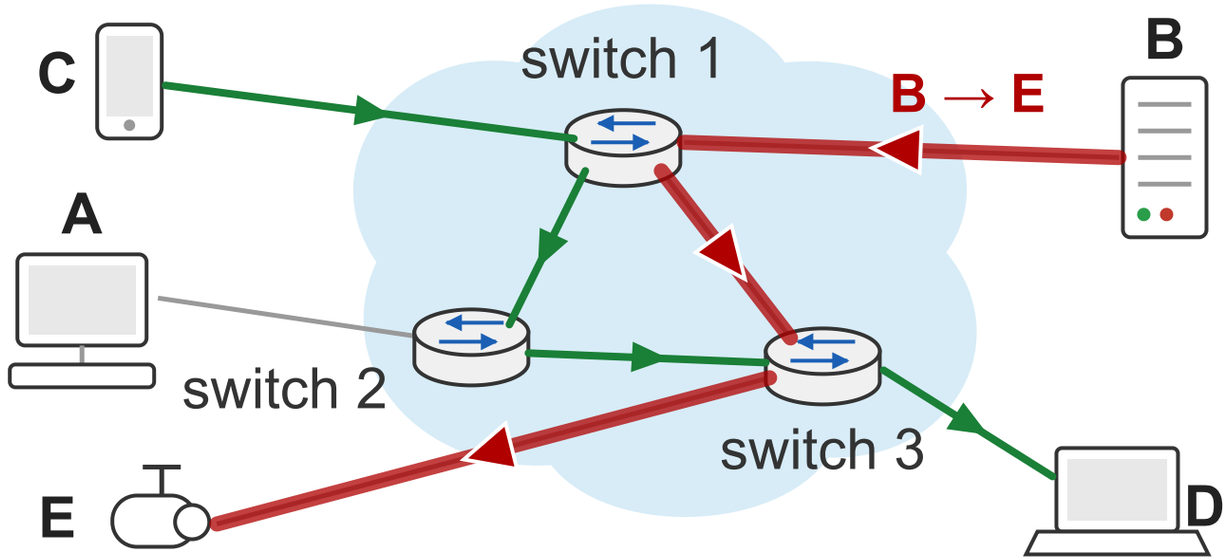
Traffic demand can change

Routing is not a one-time decision



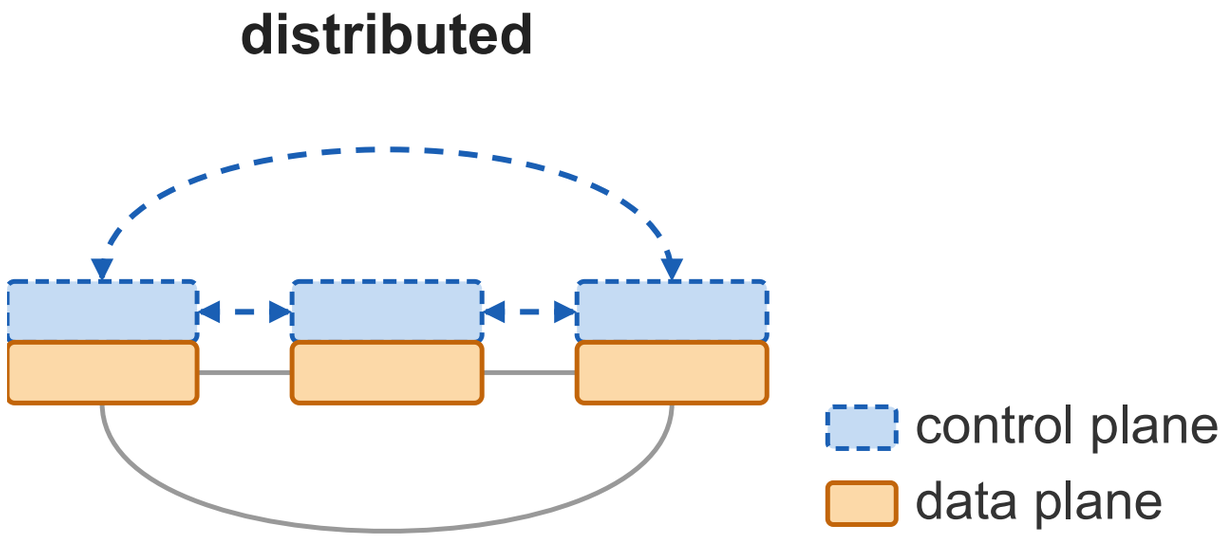
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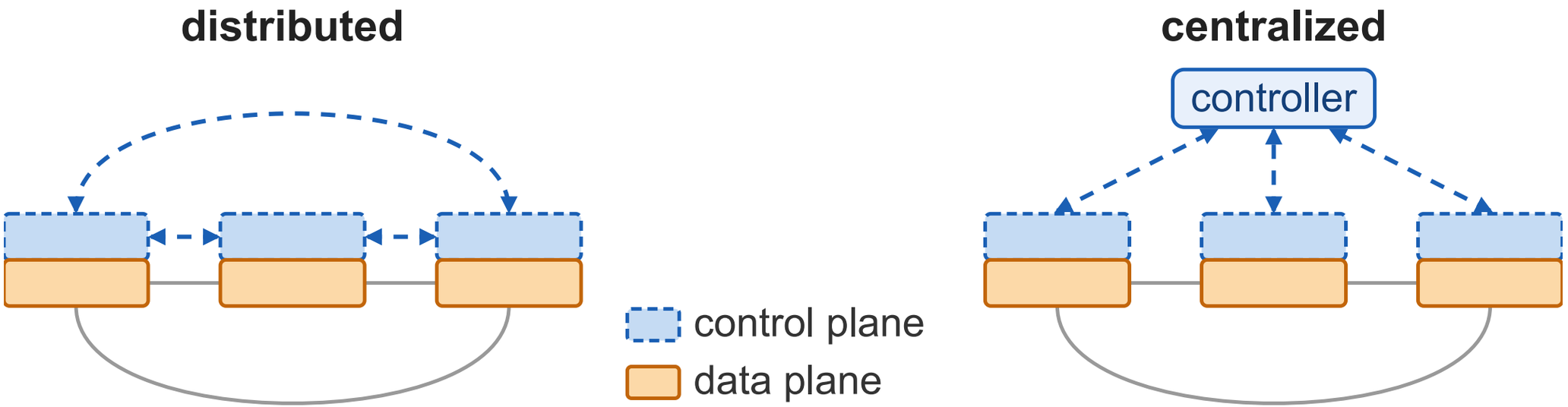
Traffic demand can change

Who decides the paths?



Switches communicate with each other to find the best paths in a distributed manner.

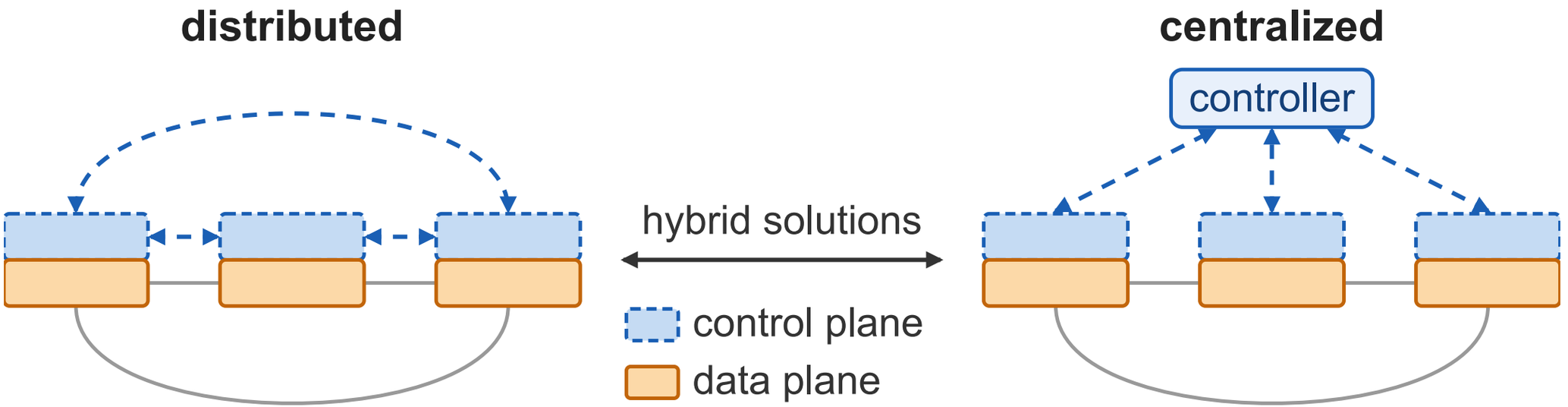
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Switches provide information to a centralized controller, who decides the paths globally, and communicates it back to the switches.

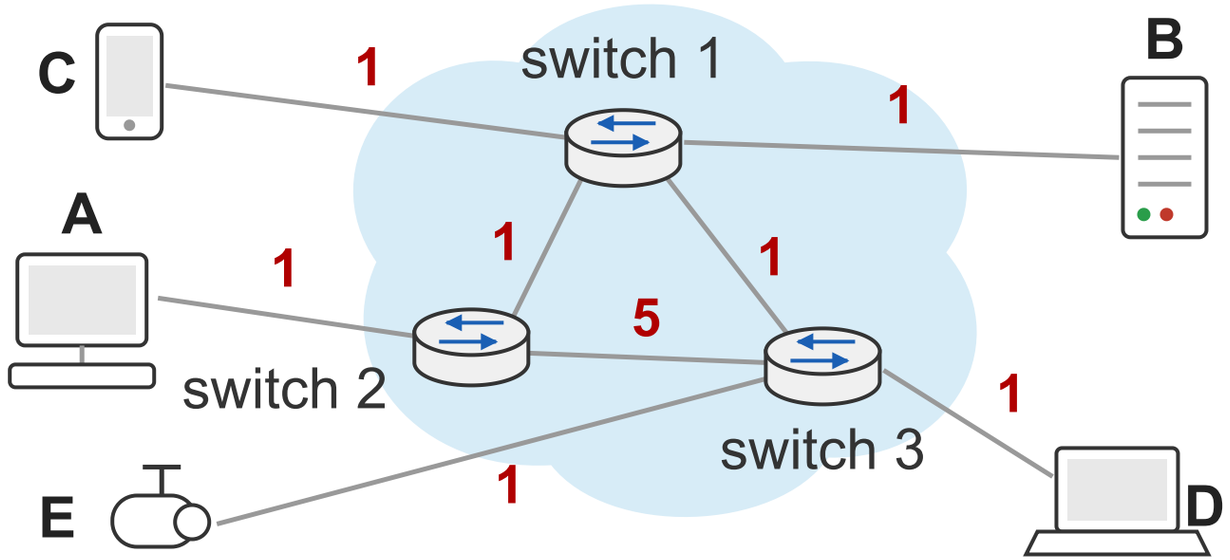
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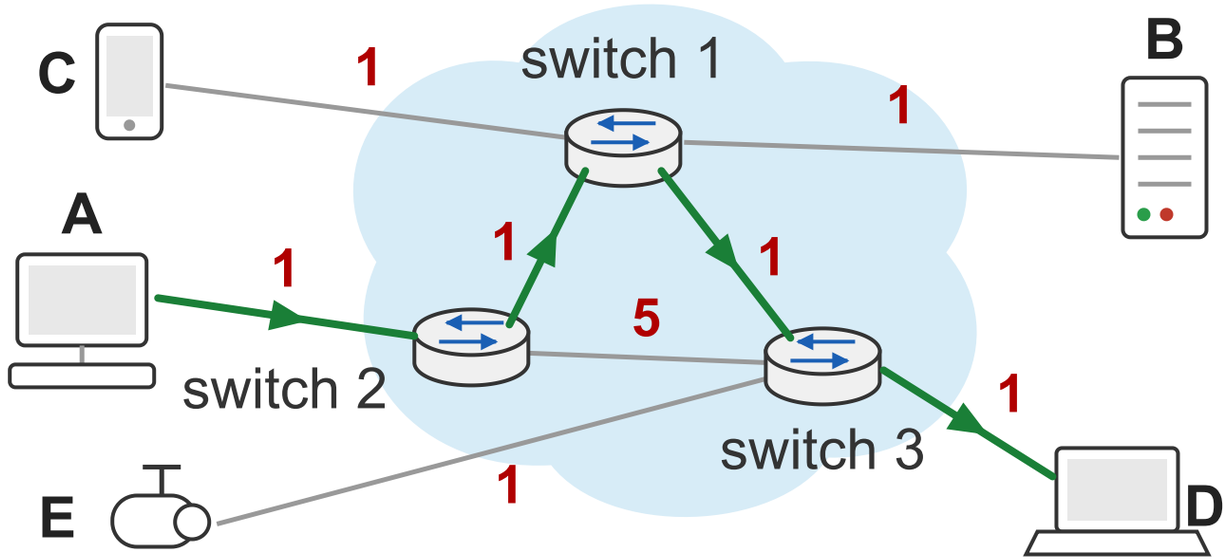
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Example policy 1: least total cost



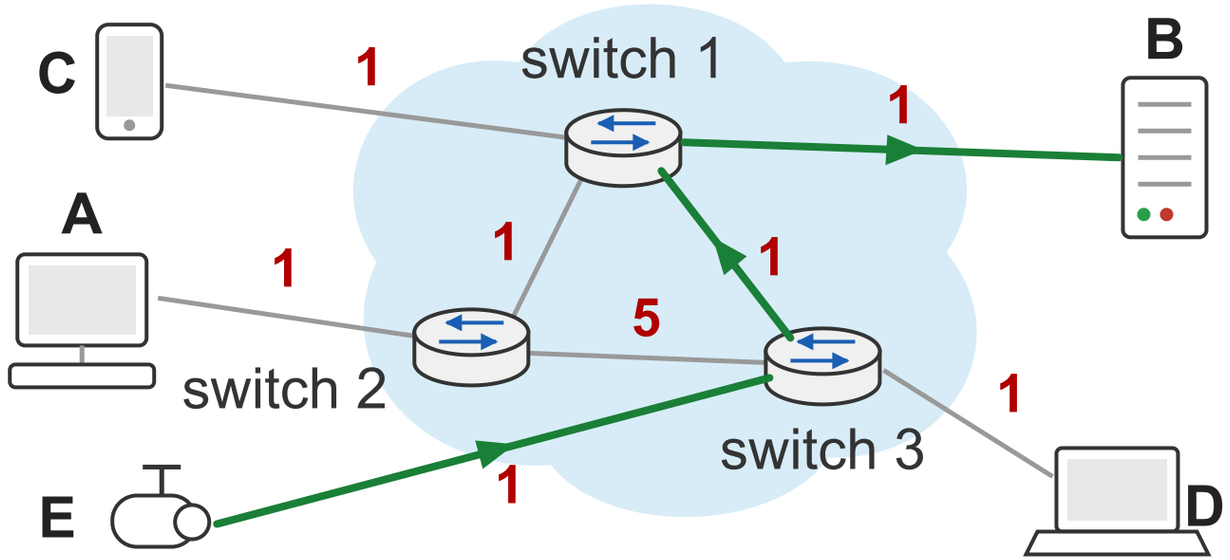
- Every link is assigned a cost, set by the network operator
- The best path between two end points is the least cost one
  - Path cost: sum of the costs of the links
- If cost is set to 1 for all links, this will boil down to the path with fewest hops

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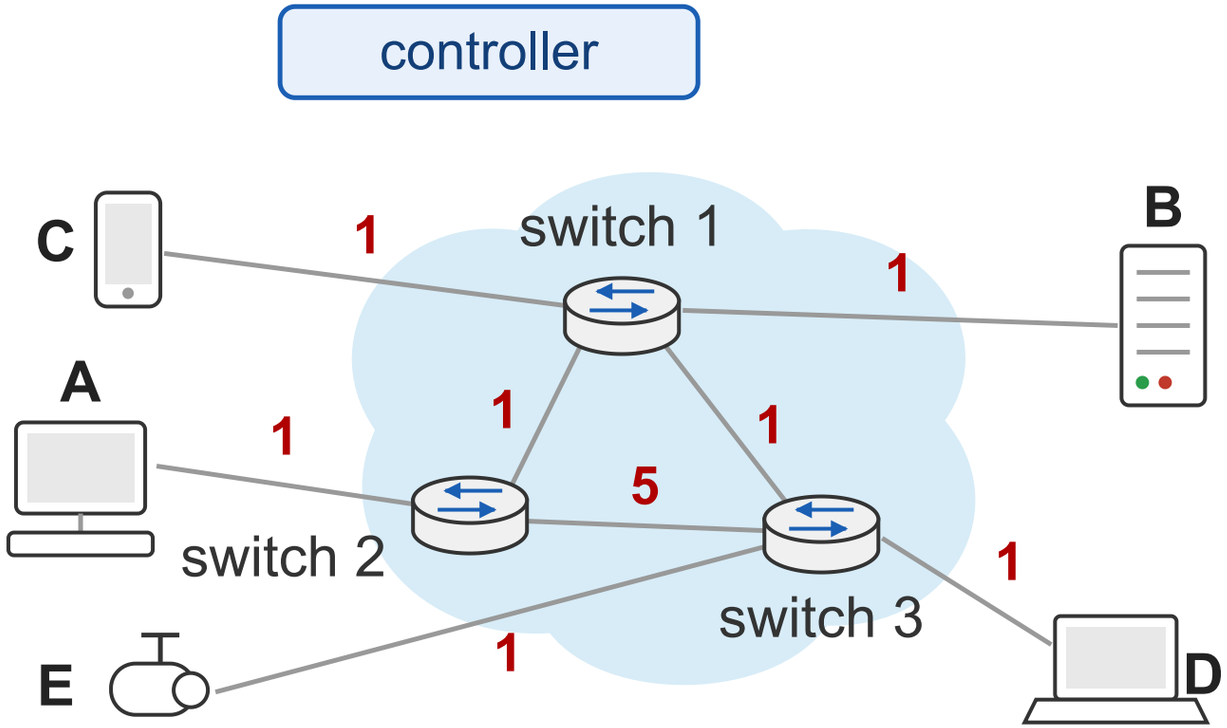
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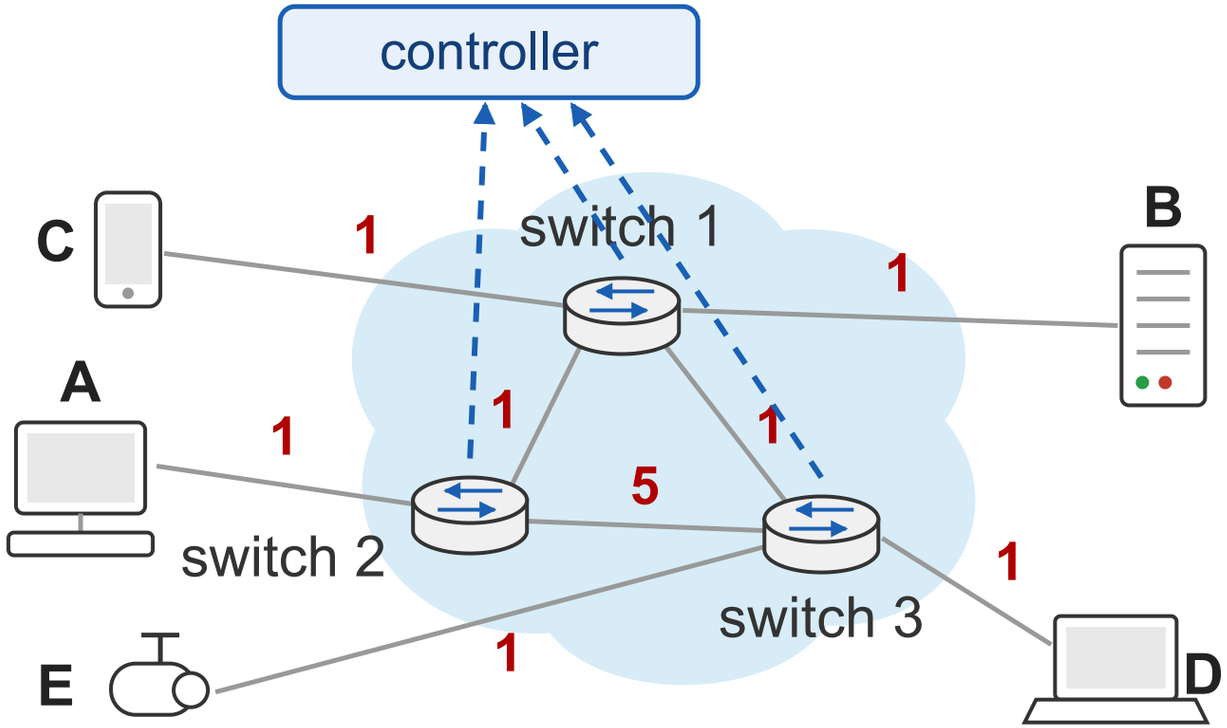


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How would this be implemented in a centralized control-plane?

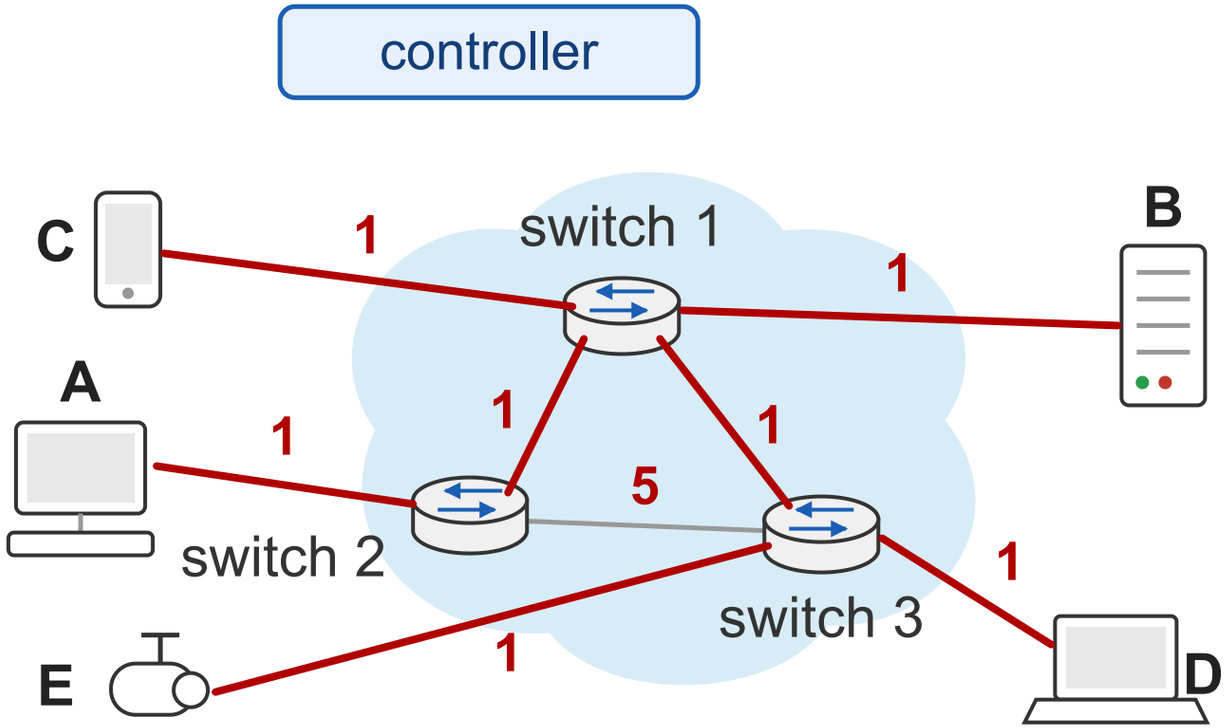


How would this be implemented in a centralized control-plane?



Every switch reports its links and their costs

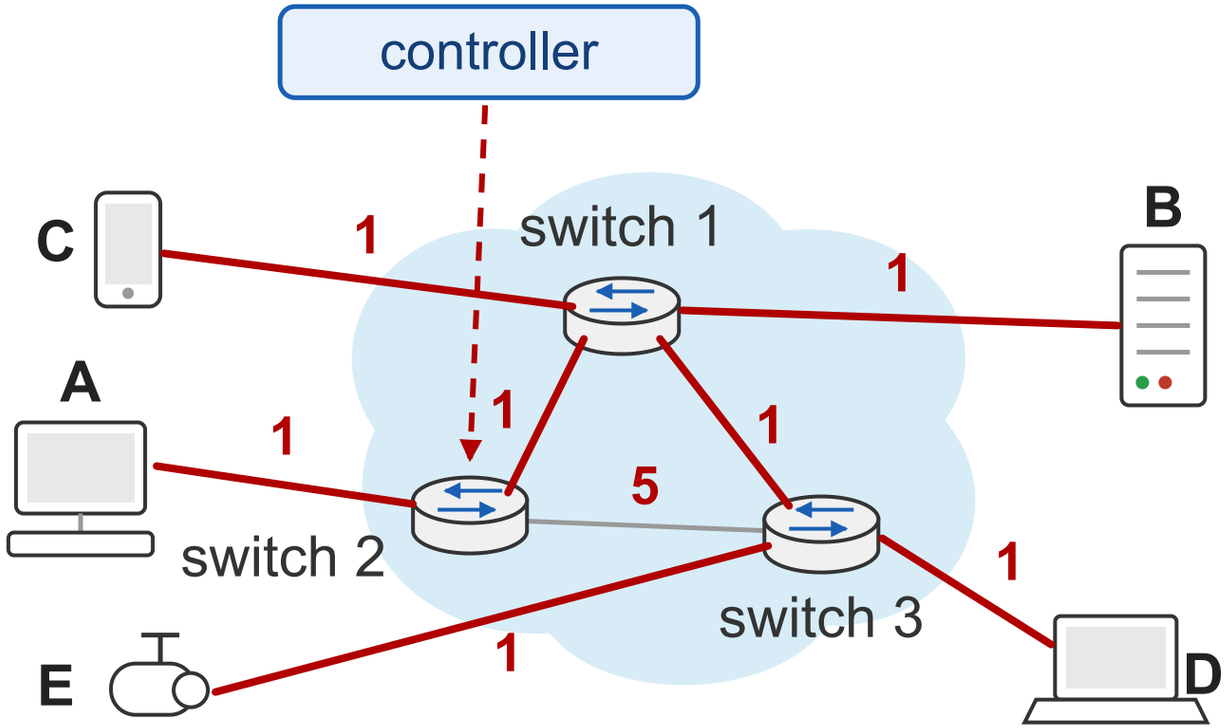
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The controller:

- finds the least-cost paths to each destination
  (e.g., using graph algorithms)

How would this be implemented in a centralized control-plane?



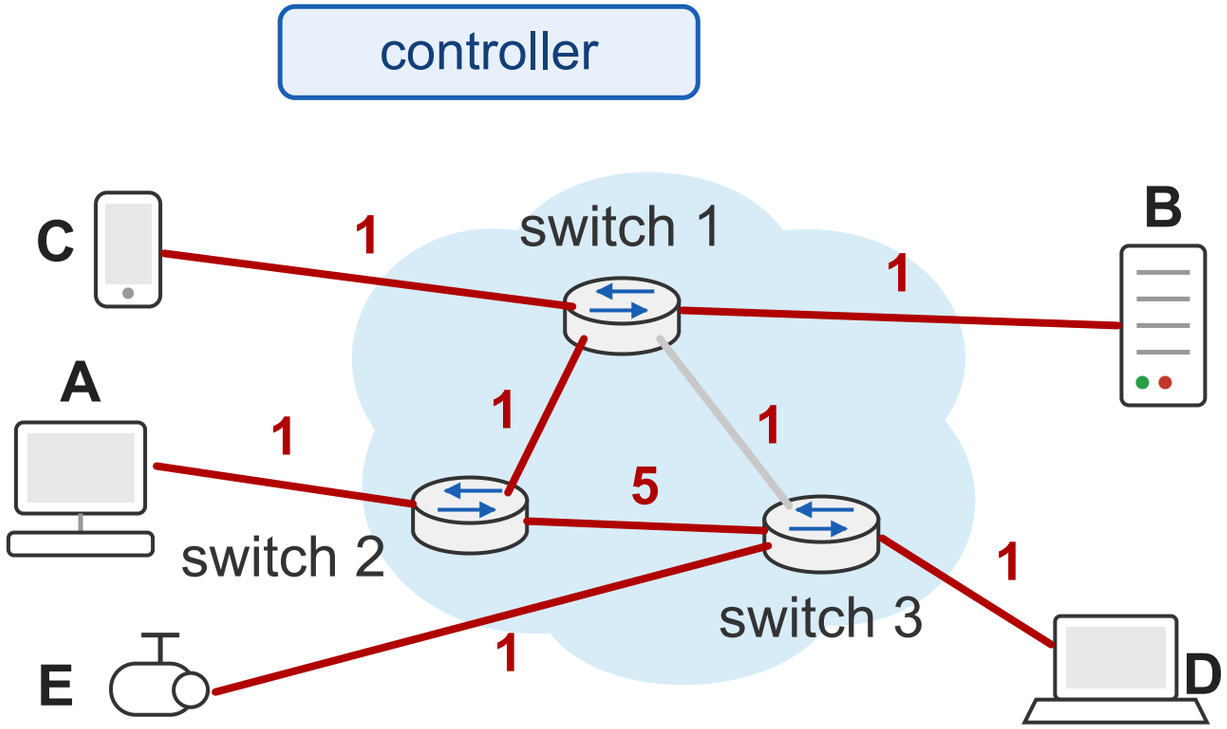
switch 2's table

destination	send on
A	to A
B	to switch 1
C	to switch 1
D	to switch 1
E	to switch 1

The controller:

- computes the forwarding table rules
- communicates the rules to the switches

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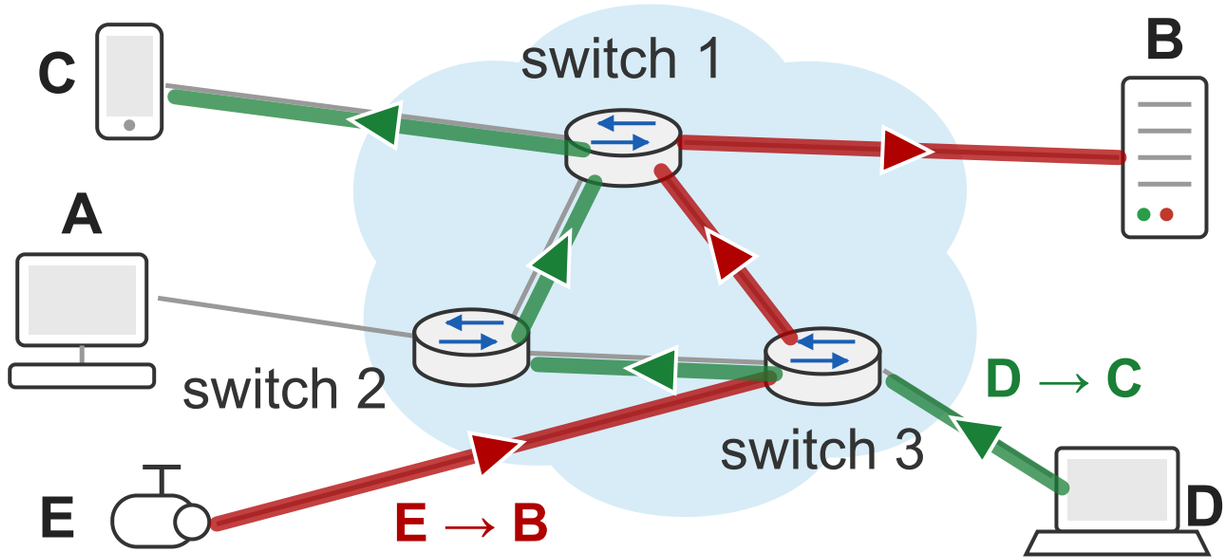


switch 2's table

destination	send on
A	to A
B	to switch 1
C	to switch 1
D	**to switch 3**
E	**to switch 3**

When the controller learns about a failure,
it modifies forwarding rules accordingly.

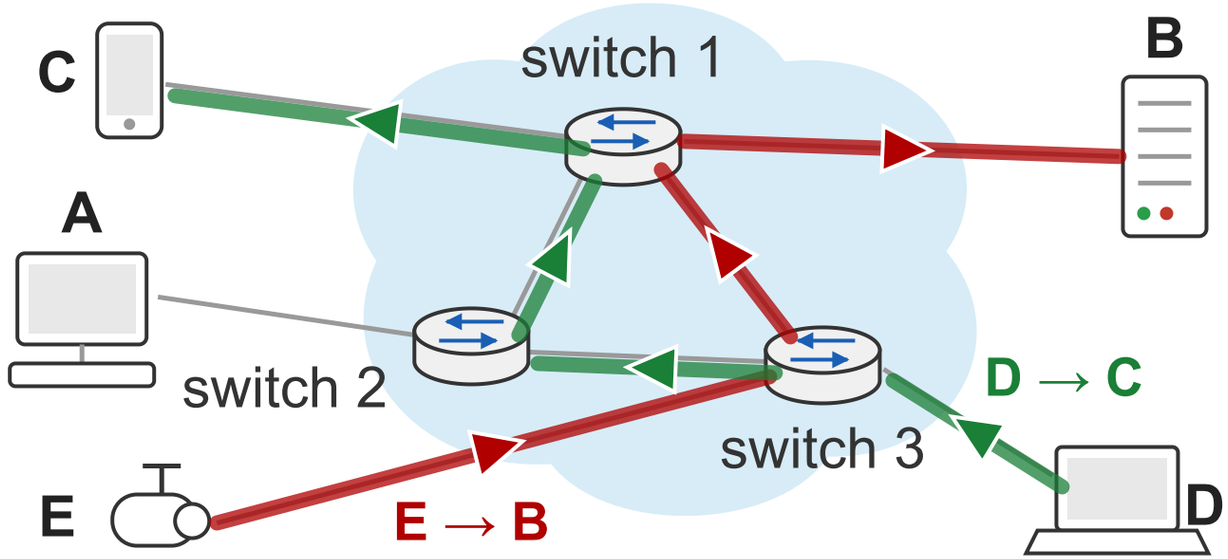
Example policy 2: keep the big flows apart



- Two long-running high-throughput traffic flows: E → B and D → C
- We don’t want their paths to overlap

Policy 2 with costs? (in-class) (in-class discussion)

Pick costs for the links so that least-cost paths keep the two flows apart.



Reset

How would this be implemented in a centralized control-plane?

